

4.7.3 Structure and role of the processor and its components

4.7.3.1 The processor and its components

Content	Additional information
Explain the role and operation of a processor and its major components: • arithmetic logic unit • control unit • clock • general-purpose registers • dedicated registers, including: • program counter • current instruction register • memory address register • memory buffer register • status register.	

4.7.3.2 The Fetch-Execute cycle and the role of registers within it

Content	Additional information
Explain how the Fetch-Execute cycle is used to execute machine code programs including the stages in the cycle (fetch, decode, execute) and details of registers used.	

4.7.3.3 The processor instruction set

Content	Additional information
Understand the term 'processor instruction set' and know that an instruction set is processor specific.	

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Know that instructions consist of an opcode and one or more operands (value, memory address or register).	A simple model will be used in which the addressing mode will be incorporated into the bits allocated to the opcode so the latter defines both the basic machine operation and the addressing mode. Students will not be expected to define opcode, only interpret opcodes in the given context of a question. For example, 4 bits have been allocated to the opcode (3 bits for basic machine operation, eg ADD, and 1 bit for the addressing mode). 4 bits have been allocated to the operand, making the instruction, opcode + operand, 8 bits in length. In this example, 16 different opcodes are possible ($2^4 = 16$). <table><tr><th colspan="4">Opcode</th><th rowspan="3">Operand</th></tr><tr><th colspan="3">Basic Machine Operation</th><th rowspan="2">Addressing Mode</th></tr><tr><th>0</th><th>0</th><th>1</th></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0 1 0 1</td></tr></table>	Opcode				Operand	Basic Machine Operation			Addressing Mode	0	0	1	0	0	1	0	0 1 0 1
Opcode				Operand														
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4.7.3.4 Addressing modes

Content	Additional information
Understand and apply immediate and direct addressing modes.	Immediate addressing: the operand is the datum. Direct addressing: the operand is the address of the datum. Address to be interpreted as meaning either main memory or register.

4.7.3.5 Machine-code/assembly language operations

Content	Additional information
Understand and apply the basic machine-code operations of: • load • add • subtract • store • branching (conditional and unconditional) • compare • logical bitwise operators (AND, OR, NOT, XOR) • logical • shift right • shift left • halt. Use the basic machine-code operations above when machine-code instructions are expressed in mnemonic form- assembly language, using immediate and direct addressing.	

4.7.3.6 Interrupts

Content	Additional information
Describe the role of interrupts and interrupt service routines (ISRs); their effect on the Fetch-Execute cycle; and the need to save the volatile environment while the interrupt is being serviced.	

4.7.3.7 Factors affecting processor performance

Content	Additional information
Explain the effect on processor performance of: • multiple cores • cache memory • clock speed • word length • address bus width • data bus width.	